

# Kugire Data Curation for the Kokinwakashu TEI/XML Data

## 1. Introduction

Kugire — phrase breaks within waka poems — are a basic structural element of waka, yet no standardized, publicly available kugire dataset exists [1]. For TEI/XML encoding of waka, each project must annotate from scratch.

atuko315/program-to-divide-31-syllable-Japanese-poem<sup>1</sup> is one of the few existing kugire resources: a program to identify kugire with a dataset for the Hyakuninissu, later applied to the Kokinwakashu and compared against [2] and [1]. The dataset, however, cannot capture the interpretive nature of kugire, which admits no single correct answer. [3] proposes two criteria — semantic independence of core sentences (necessary) and a clear grammatical break (sufficient) — both of which remain open to interpretation. A kugire dataset should therefore collect annotations from multiple sources, each with documented criteria.

This project adds kugire as an interpretive annotation layer to the TEI/XML data of the Kokinwakashu, using rule-based and translation-based methods. Keeping kugire as a separate layer preserves the annotation criteria and any competing interpretations. The data supports analysis of waka structure, enjambment, diachronic change of kugire position, and its relation to poetic style.

## 2. Data sources

### 2.1. Base data

The base data is a lexically extended version of the TEI data of the Karoku second-year manuscript of the Kokinwakashu [4]. The extended dataset maps tokenization from the Hachidaishu Vocabulary Dataset [5] onto this segmented text, forming the basis for the kugire annotation.

### 2.2. Reference data

For the rule-based method, we use the Hachidaishu Part-of-Speech Dataset [6], which provides conjugation and part-of-speech data for each token. For the translation-based method, we use the explanatory translation of the Kokinwakashu by [7], the earliest twentieth-century explanatory translation of the anthology, digitized and published on Zenodo [8].

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<sup>1</sup><https://github.com/atuko315/program-to-divide-31-syllable-Japanese-poem>

### 3. Output Data

The output is `kokin-kugire.xml`, a TEI/XML file derived from the base data by appending kugire annotations to each poem. The original text, segmentation, and all existing TEI markup are preserved without modification; the annotation layer is additive.

Each kugire position is encoded as a `<k>` element placed after the final `<seg>` child of the corresponding line element `<l>`:

```
<l n="1" xml:id="n1">
  <seg>年の内に</seg>
  <seg>春はきにけり</seg>
  <seg>ひとゝせを</seg>
  <seg>こそとやいはむ</seg>
  <seg>ことしとやいはむ</seg>
  <k n="2" source="morph" cert="high"/>
  <k n="3" source="morph" cert="low"/>
  <k n="4" source="morph" cert="high"/>
  <k n="2" source="kaneko"/>
</l>
```

Listing 1: TEI/XML encoding of kugire annotations for poem 1 of the Kokinwakashu.

The rule-based method identifies three candidate positions. The strongest (`cert="high"`) is after the second segment, where *keri* appears in the conclusive form, ending the first complete sentence. A weaker candidate (`cert="low"`) is also placed after the third segment. The LLM-assisted translation-based method, drawing on explanatory translation [7], identifies only the second segment as a kugire. The competing candidates are preserved independently.

The `@n` attribute is the 1-based segment index; `@source` is the evidence code; `@cert` (`high` / `mid` / `low`) grades morphological certainty for rule-based annotations and is omitted for translation-based ones.

Multiple `<k>` elements at the same position represent competing interpretations — from the same source at different grammatical strengths, or from different sources — and are preserved as-is. No normalization or deduplication is applied.

The `<k>` element is a project-internal tag pending a decision on the target TEI encoding.

### 4. Annotation pipeline

The annotation follows an LLM-assisted pipeline:

1. Rule-based suggestion of candidate kugire positions based on [6]
2. LLM-assisted translation-based suggestion of candidate kugire positions based on [7], [8]
3. Review and correction in a pop-up editor showing the original poem and its reference data
4. Rendering the confirmed annotation into the TEI data

The pipeline accelerates the annotation process. The per-poem review cycle reduces cognitive load and improves accuracy.

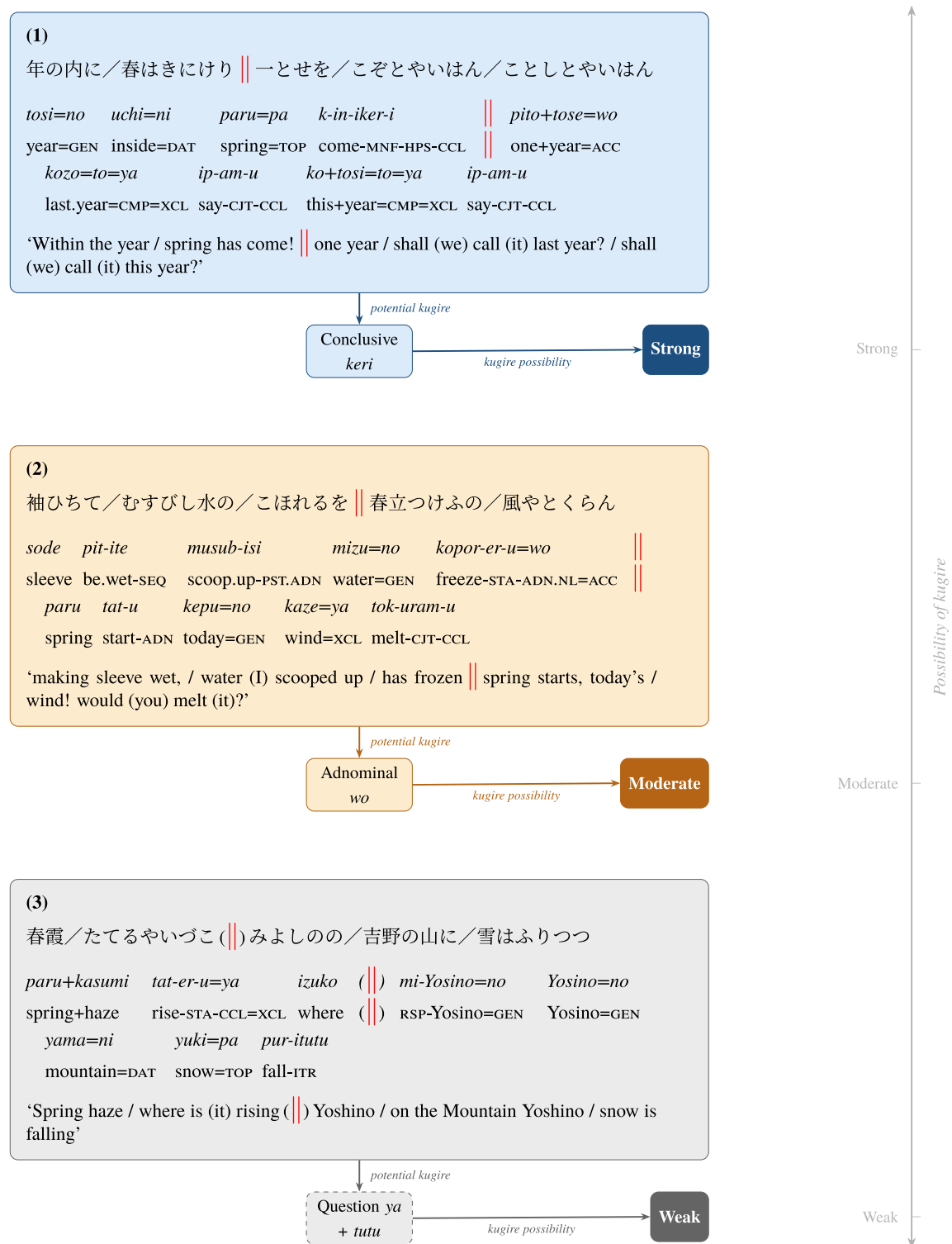
#### 4.1. Rule-based (grammar-wise) kugire suggestion

Our method extends the program<sup>2</sup>, which flags conclusive forms, copulas, and sentence-final particles as kugire, by outputting graded candidates — strong, moderate, and weak — rather than a single result.

Strong candidates are phrases ending in the conclusive form. For example, in the first poem of the *Kokinwakashu*, the past-tense auxiliary verb *keri* appears in the conclusive form, indicating a grammatical break after it (Figure 1-(1)).

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<sup>2</sup><https://github.com/atuko315/program-to-divide-31-syllable-Japanese-poem>.



Abbreviations: ACC = accusative; ADN = adnominal; ADN.NL = adnominal nominalizer; CCL = conclusive; CJT = conjunctural; CMP = complementizer; DAT = dative; GEN = genitive; HPS = hearsay past; ITR = iterative; MNF = manifestive; PST = past; RSP = respect; SEQ = sequential; STA = stative; TOP = topic; XCL = exclamative

Moderate candidates are phrases where the copula or information structure suggests a break, though not conclusively. In the second poem of the Kokinwakashu, the segments before and after *wo* are

syntactically connected, yet the former presents a scene while the latter offers a conjectural statement about it — a shift that may indicate a break (Figure 1-(2)).

Weak candidates are phrases where a sentence-final particle suggests a break, but the surrounding segments remain semantically interdependent. In the third poem of the *Kokinwakashu*, the question particle in the second segment may suggest a break, yet the following segment answers that question — semantic interdependence that makes the break uncertain (Figure 1-(3)).

## 4.2. LLM-assisted translation-based kugire suggestion

We also use a large language model (LLM) to suggest candidate kugire positions from the explanatory translation of [kaneko] as encoded by [yamamoto]. Where the rule-based approach draws on morphological and grammatical features, the LLM-assisted translation-based approach draws on the contemporary Japanese translation, capturing breaks rooted in the translator's reading rather than grammatical form alone.

The LLM is qwen2.5, served locally via Ollama. The model receives the five segments of the original poem and the corresponding explanatory translation, and outputs a JSON object with three fields: alignment (step 1), breaks (step 2), and positions (the resulting candidate indices).

The prompt instructs the model to align segments to the translation, detect sentence boundaries in the translation, and map them back to segment positions — mirroring the workflow of a human annotator using a translation as reference. Three few-shot examples from real *Kokinwakashu* poems demonstrate this chain, covering no-kugire, second-segment, and fourth-segment cases. Valid positions from the JSON output are recorded independently of rule-based results, e.g., `<k n="2" source="kaneko" />`.

## 5. Conclusion

This project produced a TEI/XML dataset prototype of kugire annotations for the *Kokinwakashu*, derived from a lexically extended base and informed by both rule-based and LLM-assisted translation-based methods. The dataset preserves multiple interpretations of kugire, supporting analysis of waka structure and diachronic change.

## 6. References

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<sup>3</sup><https://doi.org/10.5281/zenodo.13940187>

<sup>4</sup><https://doi.org/10.5281/zenodo.13942707>